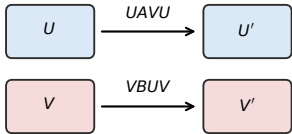


round of the layered walk

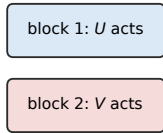


unknowns $(U, V) \in \text{Alt}(m)^2$, constants A, B

one-variable equation in $\text{Alt}(m) \wr C_2 \subseteq \text{Sym}(2m)$

block 1 = $\{0, \dots, m-1\}$, block 2 = $\{m, \dots, 2m-1\}$

Theorem
→



$$x = (U, V) \sigma$$

$$a = (B, A)$$

$$c = (U', V') \sigma$$

(U, V) solves the round $\iff x a x^2 = c$ in $\text{Alt}(m) \wr C_2$